Lecture Notes on Numerical Linear Algebra and Data Science

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Chapter 1

Linear Systems

1.1 Session 1: Matrices and Data

Matrices are a natural way to store data. A matrix

$$A = (a_{ij}) \in \mathbb{R}^{m\times n}$$

has m rows and n columns; a_{ij} denotes the entry in row i and column j. For example, a small data set could be recorded as

$$A = \begin{pmatrix} 90 & 150 & 0.2 & 1920 \\ 800 & 70 & 0.4 & 1988 \\ 150 & 90 & 1.3 & 2004 \end{pmatrix}.$$

Typically, rows represent data points and columns represent features. Thus, depending on the application, a row might represent an observation and a column its weight, age, or another measured quantity.

A vector $v \in \mathbb{R}^n$ is a matrix with one column,

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

Vectors are added componentwise and multiplied by scalars componentwise:

$$v + w = \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix}, \qquad \lambda v = \begin{pmatrix} \lambda v_1 \\ \vdots \\ \lambda v_n \end{pmatrix}.$$

Definition 1.1. *A* linear combination *of vectors $v_1, \dots, v_k$ is an expression*

$$\lambda_1 v_1 + \cdots + \lambda_k v_k, \qquad \lambda_1, \dots, \lambda_k \in \mathbb{R}.$$

For example,

$$2\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} - 3\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + 0\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \\ -14 \end{pmatrix}.$$

The coefficients may be arbitrary real numbers; a coefficient equal to zero simply means that the corresponding vector does not contribute.

For $A \in \mathbb{R}^{m\times n}$ with columns $A_1, \dots, A_n$ and $v = (v_1, \dots, v_n)^T \in \mathbb{R}^n$, matrix–vector multiplication is defined by

$$Av = v_1 A_1 + \cdots + v_n A_n.$$

For instance,

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}.$$

Equivalently, the ith entry of Av is

$$(Av)_i = \sum_{j=1}^{n} a_{ij} v_j.$$

Definition 1.2. *The* column space *of $A \in \mathbb{R}^{m \times n}$ is*

$$\mathcal{C}(A) := \{Av : v \in \mathbb{R}^n\} = \operatorname{span}\{A_1, \ldots, A_n\} \subseteq \mathbb{R}^m.$$

Geometrically, a column space in $\mathbb{R}^3$ can be only the origin, a line through the origin, a plane through the origin, or all of $\mathbb{R}^3$. Which of these possibilities occurs depends on how many genuinely different directions are present among the columns. Three columns in one plane, for instance, can generate at most that plane.

A recurring question. Given feature vectors $A_1, \ldots, A_n \in \mathbb{R}^m$ and a target vector $b \in \mathbb{R}^m$, can b be explained by the features? In algebraic form, the question is whether there are coefficients $x_1, \ldots, x_n$ such that

$$b = x_1 A_1 + \cdots + x_n A_n,$$

or, equivalently, whether $b \in \mathcal{C}(A)$. If the answer is yes, the problem is to find such coefficients. If the answer is no, we shall later look for a vector in $\mathcal{C}(A)$ that is close to b.

Adding a column which is already a linear combination of existing columns does not enlarge the column space. Equivalently, deleting a redundant column does not shrink it.

Lemma 1.3. *Let $A \in \mathbb{R}^{m \times n}$, and let A' be obtained by appending a column $A_{n+1} \in \mathcal{C}(A)$. Then*

$$\mathcal{C}(A') = \mathcal{C}(A).$$

Proof. The inclusion $\mathcal{C}(A) \subseteq \mathcal{C}(A')$ is immediate. Conversely, write $A_{n+1} = \sum_{j=1}^{n} \mu_j A_j$. Every linear combination involving the new column can then be rewritten using only $A_1, \ldots, A_n$. □

More generally, if some A_i is a linear combination of the remaining columns, then

$$\operatorname{span}\{A_1, \ldots, A_n\} = \operatorname{span}\{A_1, \ldots, A_{i-1}, A_{i+1}, \ldots, A_n\}.$$

Indeed, a coefficient multiplying A_i can be distributed among the other columns. For example, if $A_3 = A_1 + A_2$, then

$$3A_1 + 5A_2 - 7A_3 = -4A_1 - 2A_2.$$

This is why redundant features may be deleted without changing the column space.

Definition 1.4. *Vectors $v_1, \ldots, v_k \in \mathbb{R}^m$ are* linearly dependent *if one of them can be written as a linear combination of the others. Otherwise they are* linearly independent.

Equivalently, $v_1, \ldots, v_k$ are linearly independent if and only if the only solution of

$$\lambda_1 v_1 + \cdots + \lambda_k v_k = \mathbf{0}$$

is the trivial solution $\lambda_1 = \cdots = \lambda_k = 0$. Hence a matrix A has linearly independent columns precisely when $Ax = \mathbf{0}$ implies $x = \mathbf{0}$.

The criterion is often easiest to use through explicit relations. If, for example, a four-column matrix satisfies

$$A_3 = 3A_1 + 5A_2, \qquad A_4 = 4A_1 + 6A_2,$$

then its last two columns add no new directions: all four columns are generated by A_1, A_2. In contrast, when no nontrivial relation exists, the columns provide independent directions even if they live in a much larger ambient space.

Definition 1.5. *The* rank *of a matrix A is the largest number of linearly independent columns of A.*

Thus $0 \le \operatorname{rank}(A) \le n$ for $A \in \mathbb{R}^{m\times n}$. We shall soon see the stronger bound $\operatorname{rank}(A) \le \min\{m, n\}$ and an algorithm for determining the rank.

Remark 1.6. *The number of columns is the number of available features; the rank is the number of independent directions represented by them. Adding an exact copy or an exact linear combination of existing features does not increase the rank.*

1.2 Session 2: Linear Systems and Elimination

Consider a linear system

$$Ax = b, \qquad A \in \mathbb{R}^{m\times n}, \quad b \in \mathbb{R}^m,$$

where $x \in \mathbb{R}^n$ is unknown. The system asks which linear combination of the columns of A generates b. For example,

$$\begin{pmatrix} 2 & 1 & 4 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

is equivalent to

$$2x_1 + x_2 + 4x_3 = 1, \qquad x_1 + 3x_3 = -2.$$

We record the system in its augmented matrix $(A \mid b)$.

There are three elementary row operations:

1. swap two rows;
2. multiply a row by a nonzero scalar;
3. add a scalar multiple of one row to another row.

On an augmented matrix they act on the complete rows, including the right-hand side. Typical examples are

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 4 & 5 & 6 \\ 1 & 2 & 3 \end{array}\right),$$

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right) \xrightarrow{R_1 \leftarrow 2R_1} \left(\begin{array}{cc|c} 2 & 4 & 6 \\ 4 & 5 & 6 \end{array}\right), \qquad \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -3 & -6 \end{array}\right).$$

We write $A \sim B$ if B can be obtained from A by a sequence of elementary row operations. Row equivalence is symmetric because each elementary row operation has an inverse.

Lemma 1.7. *Suppose $(A \mid b) \sim (A' \mid b')$. Then x_0 solves $Ax = b$ if and only if it solves $A'x = b'$. In particular, elementary row operations preserve the solution set of a linear system.*

Proof. It is enough to check the three elementary operations. Swapping equations, multiplying an equation by a nonzero number, and replacing an equation by itself plus a multiple of another equation are all reversible and preserve exactly the same common solutions. □

Row operations generally change the column space. They do, however, preserve all linear relations among corresponding columns.

Theorem 1.8. *Let $A = (A_1, \dots, A_n)$ and $B = (B_1, \dots, B_n)$ with $A \sim B$. For every $\lambda_1, \dots, \lambda_n \in \mathbb{R}$,*

$$\lambda_1 A_1 + \cdots + \lambda_n A_n = \mathbf{0} \quad \Longleftrightarrow \quad \lambda_1 B_1 + \cdots + \lambda_n B_n = \mathbf{0}.$$

Consequently, $\operatorname{rank}(A) = \operatorname{rank}(B)$.

Proof. The coefficient vector $\lambda = (\lambda_1, \dots, \lambda_n)^T$ solves $A\lambda = \mathbf{0}$ exactly when it solves $B\lambda = \mathbf{0}$, because the two homogeneous systems are row equivalent. Thus a chosen subset of columns of A is independent if and only if the corresponding subset of columns of B is independent. □

Definition 1.9. *A matrix is in* row-echelon form *(REF) if*

1. *all nonzero rows occur above all zero rows; and*
2. *the first nonzero entry in every nonzero row occurs strictly to the right of the first nonzero entry in the row above it.*

The first nonzero entry of a nonzero row is a pivot*; its column is a* pivot column*.*

A matrix is in *reduced row-echelon form* (RREF) if, in addition, every pivot is 1 and every other entry in a pivot column is 0. Thus the pivot columns of an RREF are columns of the identity matrix. For example, a row-echelon matrix whose leading entries lie in columns $1, 3, 5$ has three pivots, irrespective of the values in its nonpivot columns.

Theorem 1.10. *Every matrix can be brought into row-echelon form, and indeed into reduced row-echelon form, by elementary row operations.*

Proof. This is the Gaussian elimination algorithm. In the leftmost column that is not yet processed, find a nonzero entry, swap its row into the current pivot position, and eliminate the entries below it. Continue on the remaining submatrix. This gives REF. Scaling pivots to 1 and eliminating above them gives RREF. □

As a larger elimination example, consider

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 12 & 1 \\ 0 & 4 & 8 & 2 & 12 & 12 \\ 0 & 8 & 16 & 4 & 6 & 21 \\ 0 & 2 & 4 & 1 & 3 & 5 \\ 0 & 6 & 12 & 3 & 21 & 13 \end{pmatrix}.$$

The first column is already zero and can never contain a pivot. In the remaining columns we first move a convenient nonzero row upward, use its leading entry to clear the entries below, and then repeat on the smaller lower-right block. Zero rows move to the bottom. The point of writing the algorithm this way is that no guessing is involved: at each stage we search for the next possible pivot and eliminate below it.

In RREF, every nonpivot column is visibly a linear combination of the pivot columns: its entries are exactly the coefficients of this combination. The pivot columns are distinct standard unit vectors and hence independent. Therefore the number of pivots is the rank. Because row operations preserve all relations between corresponding columns, the same pivot indices identify a maximal independent subset among the original columns.

Corollary 1.11. *If B is a row-echelon form of A with r pivots, then*

$$\operatorname{rank}(A) = \operatorname{rank}(B) = r.$$

In particular, for $A \in \mathbb{R}^{m\times n}$,

$$\operatorname{rank}(A) \le \min\{m, n\}.$$

In particular, any collection of more than m vectors in $\mathbb{R}^m$ is linearly dependent. Put the vectors into the columns of a matrix. Its rank is at most m, so not all of its more than m columns can be independent. This is the formal version of the geometric fact that $\mathbb{R}^m$ contains only m independent directions.

To obtain a maximal independent subset of the *original* columns of A, row-reduce A, record the pivot-column indices, and select the columns of A with precisely those indices. One must not take the columns from the reduced matrix if the goal is a basis of $\mathcal{C}(A)$, since row operations can change the column space.

1.3 Session 3: Bases, Dimension, and the Nullspace

Definition 1.12. *A nonempty subset $V \subseteq \mathbb{R}^m$ is a* linear subspace *if*

1. *$v, w \in V$ implies $v + w \in V$;*
2. *$v \in V$ and $\lambda \in \mathbb{R}$ imply $\lambda v \in V$.*

Every subspace contains $\mathbf{0}$. The most important examples for us are column spaces: for every matrix A, the set $\mathcal{C}(A)$ is a subspace.

Definition 1.13. *Vectors $v_1, \dots, v_k$* generate *a subspace V if*

$$V = \operatorname{span}\{v_1, \dots, v_k\}.$$

They form a basis *of V if they generate V and are linearly independent.*

For example, the columns of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix}$$

generate $\mathcal{C}(A)$, but they do not form a basis because $A_3 = A_1 + A_2$. The first two columns still generate the same space and form a basis. In general, a basis of $\mathcal{C}(A)$ is obtained by selecting the original columns whose indices are pivot columns in an REF of A.

Theorem 1.14. *Let $A_1, \dots, A_n$ be a basis of a subspace V, and suppose $B_1, \dots, B_k$ generate V. Then $k \geq n$.*

Proof. Since the B_j generate V, each A_i can be written as $A_i = BC_i$ for a coefficient vector $C_i \in \mathbb{R}^k$. Put $C = (C_1, \dots, C_n) \in \mathbb{R}^{k \times n}$. If $k < n$, then the columns of C are dependent, so there is a nonzero $\lambda \in \mathbb{R}^n$ with $C\lambda = \mathbf{0}$. But then

$$\lambda_1 A_1 + \cdots + \lambda_n A_n = B(C\lambda) = \mathbf{0},$$

contradicting the independence of $A_1, \dots, A_n$. □

Corollary 1.15. *Every two bases of a finite-dimensional subspace have the same number of elements.*

Definition 1.16. *The* dimension $\dim V$ *is the number of vectors in any basis of V. We set* $\dim\{\mathbf{0}\} = 0$.

It follows directly that

$$\dim \mathcal{C}(A) = \operatorname{rank}(A).$$

Definition 1.17. *For $A \in \mathbb{R}^{m \times n}$, the* nullspace *of A is*

$$\mathcal{N}(A) := \{x \in \mathbb{R}^n : Ax = \mathbf{0}\}.$$

The nullspace is a subspace of $\mathbb{R}^n$. Indeed, if $x, y \in \mathcal{N}(A)$ and $\lambda \in \mathbb{R}$, then

$$A(x + y) = Ax + Ay = \mathbf{0}, \qquad A(\lambda x) = \lambda Ax = \mathbf{0}.$$

If $A \sim B$, then $\mathcal{N}(A) = \mathcal{N}(B)$.

Example 1.18. *Consider the RREF*

$$B = \begin{pmatrix} 1 & 3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Columns 1 *and* 3 *are pivot columns; columns B_2, B_4, B_5 are called* free. *They can be expressed by the pivot columns as*

$$B_2 = 3B_1, \qquad B_4 = 0, \qquad B_5 = -B_1 + 2B_3.$$

Solving $Bx = 0$ with one free variable at a time gives the special solutions

$$s_2 = \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad s_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad s_5 = \begin{pmatrix} 1 \\ 0 \\ -2 \\ 0 \\ 1 \end{pmatrix}.$$

Every null vector is $x_2 s_2 + x_4 s_4 + x_5 s_5$. Moreover, the vectors are independent: in a linear combination of them, coordinates $2, 4, 5$ reproduce the three coefficients directly. Thus they form a basis of $\mathcal{N}(B)$, and also of $\mathcal{N}(A)$ for every $A \sim B$.

Let B be an RREF of A. After reordering variables into pivot and free variables, B has the block form

$$B = \begin{pmatrix} I_r & F \\ 0 & 0 \end{pmatrix}.$$

Writing $x = (x_P, x_F)^T$, the homogeneous system becomes

$$x_P + F x_F = 0, \qquad \text{so} \qquad x_P = -F x_F.$$

Thus

$$x = \begin{pmatrix} -F \\ I_{n-r} \end{pmatrix} x_F.$$

Setting one free variable equal to 1 and all other free variables equal to 0 gives the *special solutions*. They are independent and span the nullspace, hence form a basis of it.

Theorem 1.19 (Rank–nullity theorem)**.** *For $A \in \mathbb{R}^{m \times n}$,*

$$\dim \mathcal{N}(A) = n - \operatorname{rank}(A),$$

or equivalently,

$$\dim \mathcal{C}(A) + \dim \mathcal{N}(A) = n.$$

Proof. Let $r = \operatorname{rank}(A)$. An RREF of A has r pivot variables and $n - r$ free columns, and we get one special solution per free columns. As we said above, those are independent and span the null space, hence, $\dim \mathcal{N}(A) = n - r$. As $\dim \mathcal{C}(A) = \operatorname{rank}(A) = r$, the result follows. □

1.4 Session 4: Solving $Ax = b$ and Matrix Products

The first question about $Ax = b$ is whether the system is consistent.

Lemma 1.20. *The system $Ax = b$ has a solution if and only if the b-column is not a pivot column in an REF of the augmented matrix $(A \mid b)$.*

Proof. A pivot in the last column produces an equation of the form $0x_1 + \cdots + 0x_n = 1$, after a possible rescaling, so the system cannot have a solution. Conversely, if the last column is free, it is a linear combination of the pivot columns. Applying the same column relation to the augmented matrix shows that b is generated by the columns of A. □

Theorem 1.21. *The solution set of $Ax = b$ is either empty or of the form*

$$x_0 + \mathcal{N}(A) := \{x_0 + z : z \in \mathcal{N}(A)\},$$

where x_0 is any particular solution of $Ax = b$.

Proof. If $z \in \mathcal{N}(A)$, then $A(x_0 + z) = b$. Conversely, if x_0 and y both solve $Ax = b$, then $A(y - x_0) = b - b = \mathbf{0}$, so $y - x_0 \in \mathcal{N}(A)$. □

Unless $b = \mathbf{0}$, a nonempty solution set is generally not a subspace because it does not contain the origin. It is a translated subspace, also called an *affine subspace*. For example, let

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 4 \end{pmatrix}, \qquad b = \begin{pmatrix} 6 \\ 3 \\ 8 \end{pmatrix}.$$

Row reduction of the augmented matrix gives

$$\left(\begin{array}{ccc|c} 1 & 0 & -4/5 & 1/5 \\ 0 & 1 & 8/5 & 13/5 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Taking $x_3 = t$ therefore yields

$$x = \begin{pmatrix} 1/5 \\ 13/5 \\ 0 \end{pmatrix} + t \begin{pmatrix} 4/5 \\ -8/5 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R},$$

which geometrically is a line. Multiplying the description by 5 in the direction vector gives the equivalent form

$$x = \begin{pmatrix} 1/5 \\ 13/5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -8 \\ 5 \end{pmatrix}.$$

For which $A \in \mathbb{R}^{m\times n}$ does $Ax = b$ have a unique solution for every $b \in \mathbb{R}^m$? Uniqueness requires $\mathcal{N}(A) = \{\mathbf{0}\}$, hence $\operatorname{rank}(A) = n$. Existence for every right-hand side requires $\mathcal{C}(A) = \mathbb{R}^m$, hence $\operatorname{rank}(A) = m$. Both conditions can hold exactly when $m = n$ and $\operatorname{rank}(A) = n$.

Matrix products

Let $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$. Their product is defined column by column as

$$AB = (AB_1, AB_2, \ldots, AB_n) \in \mathbb{R}^{m\times n}.$$

Equivalently,

$$(AB)_{ij} = \sum_{\ell=1}^{k} a_{i\ell} b_{\ell j}.$$

Thus, for example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 15 & 4 \\ 33 & 7 \end{pmatrix}.$$

Whenever the dimensions match, matrix multiplication is associative and distributive:

$$(AB)C = A(BC), \qquad A(B + C) = AB + AC.$$

It is not commutative in general. Even when both products exist, usually $AB \neq BA$. The identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

satisfies $AI_n = A = I_m A$ for every $A \in \mathbb{R}^{m\times n}$.

Inverse matrices

Definition 1.22. *A square matrix $A \in \mathbb{R}^{n\times n}$ is* invertible *if there exists a matrix $B \in \mathbb{R}^{n\times n}$ such that*

$$AB = I_n.$$

Such a matrix B is called an inverse of A.

Theorem 1.23. *A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if* $\operatorname{rank}(A) = n$.

Proof. If $AB = I_n$, then every $x \in \mathbb{R}^n$ satisfies $x = ABx$, so every vector is a linear combination of the columns of A and $\operatorname{rank}(A) = n$.

Conversely, if $\operatorname{rank}(A) = n$, then each standard unit vector e_i belongs to $\mathcal{C}(A)$. Choose B_i such that $AB_i = e_i$ and put $B = (B_1, \dots, B_n)$. Then $AB = I_n$. □

Theorem 1.24. *If A is invertible, its inverse is unique. It is denoted by A^{-1} and satisfies*

$$AA^{-1} = I_n = A^{-1}A.$$

Proof. Suppose $AB = I_n = AC$. Then $A(B - C) = 0$. Since full rank implies $\mathcal{N}(A) = \{\mathbf{0}\}$, every column of $B - C$ is zero, hence $B = C$. Furthermore,

$$A(BA - I_n) = ABA - A = 0,$$

so again $BA - I_n = 0$. □

Corollary 1.25. *If A is invertible, the unique solution of $Ax = b$ is*

$$x = A^{-1}b.$$

Although this formula is useful theoretically, forming the inverse explicitly is usually too expensive and numerically less attractive than solving the linear system directly.

1.5 Session 5: LU Decomposition and a Data Example

Suppose one has to solve

$$Ax = b_i$$

for many right-hand sides $b_1, \dots, b_k$. Repeating all of Gaussian elimination for each right-hand side wastes work. The goal of a factorization is to preprocess A once and then solve each new system by cheaper triangular substitutions.

Each elementary row operation can be represented by left multiplication with an elementary matrix.

Lemma 1.26. *If A' is obtained from A by one elementary row operation, then*

$$A' = EA$$

for an invertible elementary matrix E. The matrix E is obtained by applying the same row operation to the identity matrix.

The three types of row operation are illustrated by the same matrix products as in the lecture:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 4 & 5 & 6 \end{pmatrix},$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 1 & 4 & 2 \end{pmatrix}.$$

They swap rows 2 and 3, double row 1, and subtract twice row 1 from row 2, respectively. Each operation is reversible, so its elementary matrix is invertible. Products of these matrices are invertible as well. If elimination produces $MA = U$, we therefore already have $A = M^{-1}U$. But a general invertible factor is not yet the useful triangular factor we want.

If Gaussian elimination uses only row additions below a pivot, it produces

$$MA = U,$$

where U is upper triangular (or upper trapezoidal). The inverse $L = M^{-1}$ is lower triangular with ones on its diagonal, and therefore

$$A = LU.$$

The entries below the diagonal of L record the multiples used during elimination, with the signs reversed by taking the inverse. Thus the forward solve with L replays how the right-hand side changes under elimination, without repeating the elimination of A itself. If row swaps are required, they are collected in a permutation matrix P.

Theorem 1.27 (LU decomposition with pivoting)**.** *Every matrix $A \in \mathbb{R}^{m\times n}$ can be decomposed as*

$$PA = LU,$$

where $P \in \mathbb{R}^{m\times m}$ is a permutation matrix, $L \in \mathbb{R}^{m\times m}$ is lower triangular with diagonal entries 1, and $U \in \mathbb{R}^{m\times n}$ is in row-echelon form.

To solve $Ax = b$, first solve the lower-triangular system

$$Ly = Pb$$

by forward substitution, and then the upper-triangular system

$$Ux = y$$

by backward substitution. The same factors P, L, U can be reused for every new right-hand side.

Remark 1.28. *Elimination is often presented using RREF, because RREF displays the solution set clearly. For numerical work one normally stops at an upper-triangular form. Back substitution then completes the solve without the extra work of eliminating above each pivot.*

Use case: a system from nutrition data

We want to compose a meal with prescribed amounts of protein, carbohydrates, and fat. The lecture uses the following simplified data, in grams per 100 grams of food:

	Milk (low-fat)	Soy flour	Whey
Protein	36	51	13
Carbohydrates	52	34	74
Fat	0	7	1.1

Taking x_1 grams of milk, x_2 grams of soy flour, and x_3 grams of whey yields $0.36x_1 + 0.51x_2 + 0.13x_3$ grams of protein, $0.52x_1 + 0.34x_2 + 0.74x_3$ grams of carbohydrates, and $0.07x_2 + 0.011x_3$ grams of fat. Thus the target $(33, 45, 3)^T$ gives

$$Ax = b, \qquad A = \begin{pmatrix} 0.36 & 0.51 & 0.13 \\ 0.52 & 0.34 & 0.74 \\ 0 & 0.07 & 0.011 \end{pmatrix}, \quad b = \begin{pmatrix} 33 \\ 45 \\ 3 \end{pmatrix}.$$

```
import numpy as np
A = np.array([[0.36, 0.51, 0.13],
              [0.52, 0.34, 0.74],
              [0.00, 0.07, 0.011]])
b = np.array([33.0, 45.0, 3.0])
x = np.linalg.solve(A, b)
print(x)
print(A @ np.round(x))
```

The solution is approximately

$$x = (27.7223,\ 39.1921,\ 23.3231)^T.$$

Our kitchen scale measures whole grams. Using $(28, 39, 23)^T$ gives

$$A \begin{pmatrix} 28 \\ 39 \\ 23 \end{pmatrix} = \begin{pmatrix} 32.96 \\ 44.84 \\ 2.983 \end{pmatrix},$$

which is close to the requested diet. Does rounding a solution always give an equally satisfactory result? This leads to numerical stability and conditioning, which measure the sensitivity of a calculation to perturbations.

The next day, the target changes to 26 grams of protein, 55 grams of carbohydrates, and 4 grams of fat:

$$b_2 = \begin{pmatrix} 26 \\ 55 \\ 4 \end{pmatrix}, \qquad A^{-1}b_2 \approx \begin{pmatrix} -17.7693 \\ 46.8864 \\ 65.2685 \end{pmatrix}.$$

Negative milk is not a usable ingredient! Could there be another exact solution? No: $\operatorname{rank}(A) = 3$, so A is invertible and the solution is unique. Simply discarding the negative amount and taking $(0, 47, 65)^T$ gives

$$A \begin{pmatrix} 0 \\ 47 \\ 65 \end{pmatrix} = \begin{pmatrix} 32.42 \\ 64.08 \\ 4.005 \end{pmatrix},$$

which misses the first two targets substantially. Instead, require nonnegative quantities and minimize the residual:

$$\min_{x_1, x_2, x_3 \geq 0} \|Ax - b_2\|.$$

This is the *nonnegative least-squares* (NNLS) problem.

```
from scipy.optimize import nnls
b2 = np.array([26.0, 55.0, 4.0])
print(np.linalg.solve(A, b2))
print(np.linalg.matrix_rank(A))
x_opt, residual = nnls(A, b2)
print(x_opt)
print(A @ x_opt)
```

Here $x_{\text{opt}} \approx (0, 36.5470, 57.5210)^T$ produces $(26.1167, 54.9915, 3.1910)^T$, with residual norm approximately 0.8174. It cannot satisfy every target exactly, but it is the best fit under the nonnegativity constraint and this choice of Euclidean error.

The need to minimize a residual brings us to orthogonality and least squares.

Chapter 2

Orthogonality

2.1 Session 6: Scalar Products and Norms

Linear independence distinguishes exact redundancy from independence, but it does not quantify whether two vectors point in similar directions. For data analysis we often want to measure correlation, angles, and lengths.

Definition 2.1. *For $v, w \in \mathbb{R}^n$, their* scalar product *or* inner product *is*

$$\langle v, w \rangle := v_1 w_1 + \cdots + v_n w_n.$$

The vectors are orthogonal*, written $v \perp w$, if $\langle v, w \rangle = 0$.*

The scalar product is linear in each argument, symmetric, and positive definite:

$$\begin{aligned} \langle u + v, w \rangle &= \langle u, w \rangle + \langle v, w \rangle , & \langle \lambda v, w \rangle &= \lambda \langle v, w \rangle , \\ \langle v, w \rangle &= \langle w, v \rangle , & \langle v, v \rangle &\geq 0, \end{aligned}$$

with $\langle v, v \rangle = 0$ if and only if $v = \mathbf{0}$.

Definition 2.2. *The Euclidean norm of $v \in \mathbb{R}^n$ is*

$$\|v\| := \sqrt{\langle v, v \rangle}.$$

A vector with $\|v\| = 1$ is a unit vector*. For $v \neq \mathbf{0}$, the vector $v/\|v\|$ is the normalization of v.*

The norm is the Euclidean length of a vector, and the distance between points $x, y \in \mathbb{R}^n$ is $\|x - y\|$.

The scalar product also encodes the angle between two vectors. The law of cosines applied to the triangle with sides $v, w, v - w$ gives

$$\|v - w\|^2 = \|v\|^2 + \|w\|^2 - 2\|v\| \, \|w\| \cos \alpha.$$

On the other hand, direct algebra gives

$$\|v - w\|^2 = \langle v - w, v - w \rangle = \|v\|^2 + \|w\|^2 - 2 \langle v, w \rangle .$$

Comparing the two expressions yields

$$\langle v, w \rangle = \|v\| \, \|w\| \cos \alpha.$$

In particular, $\langle v, w \rangle = 0$ corresponds to a right angle.

Theorem 2.3 (Cauchy–Schwarz inequality)**.** *For all $v, w \in \mathbb{R}^n$,*

$$| \langle v, w \rangle | \leq \|v\| \, \|w\|.$$

Equality holds if and only if v and w are linearly dependent.

For nonzero v, w, Cauchy–Schwarz implies that

$$\cos\alpha = \frac{\langle v, w\rangle}{\|v\|\,\|w\|}$$

lies in $[-1, 1]$. This equation defines the angle α between the two vectors. The normalized scalar product is often used as a similarity or correlation measure.

Theorem 2.4 (Pythagorean theorem)**.** *If $v \perp w$, then*

$$\|v + w\|^2 = \|v\|^2 + \|w\|^2.$$

Proof. Expand

$$\|v + w\|^2 = \langle v + w, v + w\rangle = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$$

and use $\langle v, w\rangle = 0$. □

Theorem 2.5 (Triangle inequality)**.** *For all $v, w \in \mathbb{R}^n$,*

$$\|v + w\| \le \|v\| + \|w\|.$$

Proof. Cauchy–Schwarz gives

$$\|v + w\|^2 = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2 \le (\|v\| + \|w\|)^2.$$

□

Transpose matrices

For $A \in \mathbb{R}^{m\times n}$, the transpose $A^T \in \mathbb{R}^{n\times m}$ is defined by

$$(A^T)_{ij} = A_{ji}.$$

For example,

$$A = \begin{pmatrix} 1 & 2 \\ 1 & -1 \\ 3 & 4 \end{pmatrix} \quad\Longrightarrow\quad A^T = \begin{pmatrix} 1 & 1 & 3 \\ 2 & -1 & 4 \end{pmatrix}.$$

Thus the ith row of A^T is the ith column of A. The basic identities are

$$(A^T)^T = A, \qquad (A + B)^T = A^T + B^T, \qquad (AB)^T = B^T A^T.$$

For the product identity, if $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$, compare entries:

$$((AB)^T)_{ij} = (AB)_{ji} = \sum_{\ell=1}^{k} a_{j\ell} b_{\ell i} = \sum_{\ell=1}^{k} (B^T)_{i\ell}(A^T)_{\ell j} = (B^T A^T)_{ij}.$$

Since $\langle v, w\rangle = v^T w$, transpose notation connects inner products to matrix multiplication.

Lemma 2.6. *For $A \in \mathbb{R}^{m\times n}$, $v \in \mathbb{R}^n$, and $w \in \mathbb{R}^m$,*

$$\langle Av, w\rangle = \left\langle v, A^T w\right\rangle.$$

Proof.

$$\langle Av, w\rangle = (Av)^T w = v^T A^T w = \left\langle v, A^T w\right\rangle.$$

□

The vector $A^T v$ contains the scalar products of v with the columns of A:

$$A^T v = \begin{pmatrix} \langle A_1, v\rangle \\ \vdots \\ \langle A_n, v\rangle \end{pmatrix}.$$

The Gram matrix

$$A^T A = (\langle A_i, A_j\rangle)_{i,j=1}^n$$

is square and symmetric. It records pairwise correlations among the columns of A. For a tall data matrix, $A^T A$ can be much smaller than A, although forming it may discard information and can magnify numerical conditioning problems.

2.2 Session 7: Projections and Orthogonal Bases

Projection onto a line

Let $w \in \mathbb{R}^n$ be a unit vector, and let $L = \operatorname{span}\{w\}$. The point on L closest to a given $v \in \mathbb{R}^n$ is

$$\widehat{v} = \langle v, w\rangle\, w.$$

Indeed,

$$\langle v - \widehat{v}, w\rangle = \langle v, w\rangle - \langle v, w\rangle \langle w, w\rangle = 0,$$

so $v - \widehat{v}$ is orthogonal to the line. For every $q \in L$,

$$v - q = (v - \widehat{v}) + (\widehat{v} - q)$$

is a sum of orthogonal vectors. Therefore

$$\|v - q\|^2 = \|v - \widehat{v}\|^2 + \|\widehat{v} - q\|^2 \geq \|v - \widehat{v}\|^2.$$

If w is not normalized, the formula becomes

$$\widehat{v} = \frac{\langle v, w\rangle}{\|w\|^2} w.$$

If w_1, w_2 are orthonormal, then projection onto the plane they span is

$$\widehat{v} = \langle v, w_1\rangle\, w_1 + \langle v, w_2\rangle\, w_2.$$

The cross terms vanish precisely because the spanning vectors are orthogonal. If the spanning vectors are not orthonormal, these two coefficients are in general wrong; one must first orthonormalize the spanning vectors or solve the resulting system for the projection coefficients.

Definition 2.7. *A basis $b_1, \ldots, b_m$ of a subspace $V \subseteq \mathbb{R}^n$ is* orthogonal *if*

$$b_i \perp b_j \qquad (i \neq j).$$

It is orthonormal *if, in addition, $\|b_i\| = 1$ for every i.*

An orthogonal basis can always be normalized by replacing each b_i with $b_i/\|b_i\|$. The standard vectors $e_1, \ldots, e_n$ form an orthonormal basis of $\mathbb{R}^n$.

For a less trivial example, consider

$$b_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad b_2 = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}, \qquad b_3 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.$$

All pairwise scalar products are zero, and their norms are respectively $\sqrt{2}, \sqrt{3}, \sqrt{6}$. Hence

$$\frac{b_1}{\sqrt{2}}, \qquad \frac{b_2}{\sqrt{3}}, \qquad \frac{b_3}{\sqrt{6}}$$

form an orthonormal basis of $\mathbb{R}^3$.

Another example starts with any basis v, w of $\mathbb{R}^2$, where $\|w\| = 1$. Then w and $v - \langle v, w\rangle\, w$ form an orthogonal basis: the second vector is perpendicular to w and is nonzero by independence.

Why choose an orthogonal basis? With a general basis, finding the coefficients of v requires solving $(b_1, \dots, b_m)\lambda = v$. With an orthogonal basis, scalar products give the coefficients directly.

Lemma 2.8. *Every list of nonzero, pairwise orthogonal vectors is linearly independent.*

Proof. Suppose $\lambda_1 b_1 + \cdots + \lambda_m b_m = \mathbf{0}$. Taking the inner product with b_i gives

$$0 = \lambda_i \|b_i\|^2,$$

because all other terms vanish. Thus every $\lambda_i = 0$. □

The projection formula also supplies the missing proof of Cauchy–Schwarz. For $w \neq 0$, the squared norm of the part of v perpendicular to w is nonnegative:

$$\begin{aligned} 0 &\le \left\| v - \frac{\langle v, w\rangle}{\|w\|^2} w \right\|^2 \\ &= \|v\|^2 - \frac{\langle v, w\rangle^2}{\|w\|^2}. \end{aligned}$$

Multiplication by $\|w\|^2$ gives $\langle v, w\rangle^2 \le \|v\|^2 \|w\|^2$, and taking square roots proves the inequality. Equality holds precisely when the perpendicular part is zero, that is, when v is a multiple of w. The case $w = 0$ is immediate.

Theorem 2.9. *If $b_1, \dots, b_m$ is an orthogonal basis of V and $v \in V$, then*

$$v = \sum_{i=1}^{m} \frac{\langle v, b_i\rangle}{\|b_i\|^2} b_i.$$

For an orthonormal basis this simplifies to

$$v = \sum_{i=1}^{m} \langle v, b_i\rangle\, b_i.$$

Proof. Write $v = \sum_i \lambda_i b_i$ and take the inner product with b_j. All terms but one vanish, giving $\langle v, b_j\rangle = \lambda_j \|b_j\|^2$. □

Definition 2.10. *For a subspace $V \subseteq \mathbb{R}^n$, its* orthogonal complement *is*

$$V^{\perp} := \{w \in \mathbb{R}^n : \langle w, v\rangle = 0 \text{ for all } v \in V\}.$$

The set $V^{\perp}$ is itself a subspace. If $b_1, \dots, b_m$ is an orthonormal basis of V, define

$$\widehat{v} := \sum_{i=1}^{m} \langle v, b_i\rangle\, b_i.$$

Then $\widehat{v} \in V$ and $v - \widehat{v} \in V^{\perp}$.

Theorem 2.11 (Best approximation theorem)**.** *Let $V \subseteq \mathbb{R}^n$ be a subspace with orthonormal basis $b_1, \dots, b_m$. For every $v \in \mathbb{R}^n$, the vector*

$$\widehat{v} = \sum_{i=1}^{m} \langle v, b_i \rangle \, b_i$$

is the unique solution of

$$\min_{q \in V} \|v - q\|.$$

Proof. We have $v - \widehat{v} \in V^\perp$. For $q \in V$, also $q - \widehat{v} \in V$, so

$$q - v = (q - \widehat{v}) + (\widehat{v} - v)$$

is an orthogonal decomposition. Hence

$$\|q - v\|^2 = \|q - \widehat{v}\|^2 + \|\widehat{v} - v\|^2 \geq \|\widehat{v} - v\|^2.$$

Equality holds only when $q = \widehat{v}$. □

Projection matrices

Put the orthonormal basis vectors into the columns of $B = (b_1, \dots, b_m) \in \mathbb{R}^{n \times m}$. Then

$$B^T B = I_m, \qquad \widehat{v} = BB^T v.$$

Thus the orthogonal projector onto V is

$$\Pi_V = BB^T.$$

It is symmetric and idempotent:

$$\Pi_V^T = \Pi_V, \qquad \Pi_V^2 = \Pi_V.$$

The second identity expresses the geometric fact that projecting a vector already in V does nothing.

Example 2.12 (Projection onto a line in $\mathbb{R}^4$)**.** *Take the unit vector*

$$b_1 = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}, \qquad B = (b_1).$$

The projection matrix and the projection of the first coordinate vector are

$$BB^T = \frac{1}{4} \begin{pmatrix} 1 & -1 & 1 & 1 \\ -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{pmatrix}, \qquad BB^T \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}.$$